

Lab 3 — Architectural Pattern Selection

Self-Service Coffee Kiosk System

Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice: The system is organized into Presentation, Business, and Data/Hardware layers. The Customer UI handles touch-screen interaction; the Order Manager and Payment Service handle business operations; the Menu Database stores menu and pricing data; and the Receipt Printer provides hardware output.

Reason 1: Clear separation of responsibilities: customer interaction is separated from order/payment processing and data storage, making the design easier to understand and maintain.

Reason 2: The architecture fits the kiosk interactions: the Order Manager coordinates menu data, credit-card payment processing, and receipt printing without exposing the internal implementation of those components.

Security Advantage: The user interface communicates with the business layer rather than directly accessing payment processing or stored data, allowing validation and access controls to be applied at the business layer.

Performance Benefit: Menu and pricing information can be retrieved directly by the Order Manager from the Menu Database, while the separate components keep the ordering flow focused on the required kiosk interactions.